

Literal Capture in Typed Decision Models: A Behavioural Audit of Calibration, Option-Order Stability and Figurative Meaning in Bangla Idioms

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Abstract

Decision models, non-autoregressive scorers that return a calibrated probability over caller-supplied options and emit no text, are increasingly deployed as routers, classifiers and guardrails. They generate nothing, so the usual explainability toolkit of rationales, chain-of-thought and token attribution does not apply, and idioms make token-level attribution ill-posed in principle: a non-compositional expression has no word whose removal is informative. We therefore audit three such models, an encoder scorer, a decoder scorer and a closed commercial API, entirely through *interventions on the option set*, using 10,361 expert-annotated Bangla idioms that ship with both a gold figurative gloss and a gold literal gloss. Our instrument holds the item, the instruction and k fixed and varies only the distractor family, which isolates one failure mode at a time and needs no new annotation. One result holds across all three arms: when the options are stripped of semantics, so that accuracy is pinned at chance by construction, reported confidence ranges from 0.28 to 0.82, and the arm that is best calibrated on real options is the worst here, returning 0.815 confidence at 0.254 accuracy. Two further results hold across the two open arms, which share a language and a paired item set. Inserting an idiom’s own literal gloss drives both well below chance, and the two trade places: the arm that is better on every other condition is the more captured one. Confidence inverts under the same shift, so confidence-gated selective prediction does worse than random gating. Option-order sensitivity, by contrast, separates the two scoring architectures rather than the two models: 83.7% against 10.7% of items change their chosen gloss across the 24 orderings of the same four options.

1 Introduction

A *decision model* is given a question and a list of candidate options and returns a probability distri-

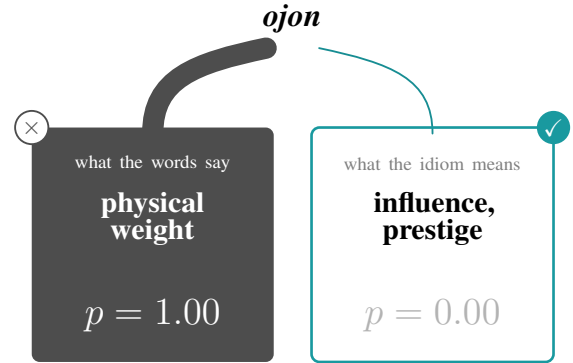


Figure 1: **Literal capture.** *Ojon* means physical weight literally and influence or prestige figuratively, much as English *carry weight* does. Offered both expert glosses among four options, Laya-ml puts all of its probability on the words and none on the idiom. Across the corpus the same swap moves it from correct to the literal reading on 64% of the items it had answered correctly, and Lod-lille, the stronger of the two on every other condition, on 88%.

bution over exactly those options. It is not autoregressive, it does not sample, and it produces no text. This class has grown quickly as infrastructure for routing, intent classification, content gating and tool selection, precisely because it is cheap, bounded and returns a number a caller can threshold on. An ecosystem analysis of 2,170 public projects finds it already embedded as a reusable decision component across domains (Ling et al., 2026), and proposals now place these models at the adjudication points of autonomous security harnesses (dos Santos, 2026). The threshold is load-bearing.

That bounded interface is also an explainability problem. There is no rationale to inspect, no chain of thought to measure for faithfulness (Jacovi and Goldberg, 2020; Turpin et al., 2023), and in the commercial case no weights at all. What is left is the behavioural surface: the caller controls the question and the option list, and sees a probability

vector. Any audit of this class has to be built out of that.

The obvious fallback, attributing the decision to input tokens (Ribeiro et al., 2016; Lundberg and Lee, 2017), is not merely unreliable here (Jain and Wallace, 2019; Adebayo et al., 2018; Hooker et al., 2019); for our domain it is ill-posed. An idiom is non-compositional by definition (Nunberg et al., 1994): the words of *kick the bucket* do not carry fractions of its meaning, so a Shapley value over them attributes a quantity that does not exist. The expression has to be intervened on as a unit, or not at all.

We take the option set as the intervention surface instead. Bangla *bāgdhārā* (idioms) are well suited to this because the expert resource we use ships, for each idiom, both a gold figurative meaning and a gold *literal* gloss: a ready-made, human-certified counterfactual distractor that no annotation effort of ours produced. Dropping that gloss into the option list turns a benchmark question into a targeted probe: does the model read the expression as an idiom, or as its words? Figure 1 shows one item answered both ways.

We contribute (i) a meaning-choice instrument for decision models in which the *distractor family* is the independent variable and everything else is held fixed, requiring no new annotation (§4); (ii) the Literal Attraction Rate, and with it a crossover interaction between the two open arms, in which the arm that is better on every non-literal condition is the more captured one (§5.1); (iii) evidence that confidence is anti-diagnostic under this shift across three different calibration philosophies, with confidence-gated selective prediction worse than random gating (§5.2); (iv) a clean architectural dissociation in option-order dependence, with a methodological consequence for anyone evaluating this class (§5.4); and (v) a semantics-free control, the one condition that runs identically on all three arms, bounding what each model reports when the options provably carry no information (§5.3).

2 Related Work

The decision-model class. The evaluation literature is still forming: an audit of 28 papers from the first nine days after release finds the typed read-out shows no independent accuracy advantage, the gains sitting in latency, cost and cascade placement (Tang and Zheng, 2026). Independent evaluations agree on the shape, reporting competitive quality at

far lower cost across rubric judging, social-science annotation, alignment-failure detection and shared-context batching (Rao and Callison-Burch, 2026; Ibrahim and Zaki, 2026; Guo et al., 2026; Yu and Yao, 2026).

Two results bear directly on ours. Ibrahim and Zaki (2026) report decision-model *calibration exceeding most LLM baselines* and recommend a cheap first stage that routes uncertain items onward; Li et al. (2026) build that system, accepting high-confidence verdicts and escalating the rest. Both are load-bearing on the returned confidence being diagnostic, and §5.2–§5.3 give the conditions under which it is not, a boundary condition on the pattern rather than a contradiction of it. Concurrently, Sun et al. (2026) show these models follow an option’s *name* rather than the definition bound to it, type-safety masking the failure because every output stays well-formed, and Xu (2026) flip 61.4% of correct decisions with benign context. §5.5 and §5.6 meet both from the opposite direction.

Idioms in NLP. Idiom resources and tasks are well established (Haagsma et al., 2020; Tayyar Madabushi et al., 2022; Chakrabarty et al., 2022), including for Bangla (Sakhawat et al., 2026), and the difficulty of figurative language for distributional models is long-noted (Shwartz and Dagan, 2019). Psycholinguistics frames the phenomenon as competition between a literal and a figurative route whose resolution depends on decomposability and context (Cacciari and Tabossi, 1988; Gibbs et al., 1989; Titone and Connine, 1999), and Oh et al. (2026) give mechanistic evidence that the competition is real inside a transformer: causal tracing recovers two pathways carrying the figurative and literal readings simultaneously, with early layers promoting the figurative one and a direct route keeping the literal one available.

Our D-LIT intervention is the behavioural counterpart of that picture rather than a replication of it. Oh et al. (2026) open an autoregressive model with full weight access and locate where the two readings live; we place both readings in the option list of a model that emits no text and, in one arm, exposes no weights at all. Their account makes our outcome measure interpretable but cannot be run on a closed decision API, which is the setting here: they can ask *how* the literal route stays available, we can only ask *how often* it wins.

Evaluating option scorers. Option-order sensitivity and selection bias are documented for

prompted generative LMs (Pezeshkpour and Hruschka, 2024; Wei et al., 2024; Zheng et al., 2024; Zhao et al., 2021). We are not aware of a comparison of option-marker against independent-option scoring in small non-autoregressive models over the full permutation set, which is what §5.4 provides.

Calibration and selective prediction. Proper scoring and calibration measurement are standard (Brier, 1950; Gneiting and Raftery, 2007; Guo et al., 2017; Nixon et al., 2019; Kumar et al., 2019), as is the observation that calibration degrades under distribution shift (Ovadia et al., 2019; Jiang et al., 2021). Selective prediction gives the operational test (Geifman and El-Yaniv, 2017, 2019; Traub et al., 2024). We report the combination that matters for deployment: accuracy *decreasing* with confidence, and a confidence gate underperforming a random one.

Behavioural testing. Our design follows behavioural checklists, contrast sets and counterfactual minimal pairs (Ribeiro et al., 2020; Gardner et al., 2020; Kaushik et al., 2020; Naik et al., 2018; Jia and Liang, 2017); the contribution is carrying that tradition into a model class with no text output, where it is one of the few instruments left.

Low-resource evaluation. Bangla has strong encoders and corpora (Bhattacharjee et al., 2022; Kakwani et al., 2020), and cross-lingual benchmarks establish the transfer gap (Hu et al., 2020; Ruder et al., 2021; Adelani et al., 2021); §5.7 localises our failure to the Bangla side rather than to figurativeness.

3 Dataset

Source corpus. We use *Bangla Bagdhara* (Sakhawat et al., 2026), a corpus of 10,361 Bangla (Bengali) idioms annotated under a 19-field schema by expert consensus.¹ Each row carries the idiom, a gold figurative meaning in Bangla, an expert literal gloss, semantic tags, alternative surface forms certified as meaning-preserving, and a gold carrier sentence. We add no annotation; every intervention below is assembled from fields that already exist.

Its own benchmark on the same underlying task reports no model among 30 multilingual and instruction-tuned LLMs above 50% accuracy,

¹Two rows are lost to parsing, leaving 10,359; of these 8,876 carry every field the instrument requires.

against 83.4% for human annotators. Option construction differs, so those figures are not directly comparable to ours, but they locate the difficulty in the material rather than in our instrument.

4 Methodology

Figure 2 sets out the instrument end to end: what the corpus supplies, what each condition changes, and what comes back.

4.1 The meaning-choice task

Given an idiom, choose its gold figurative meaning from $k = 4$ candidates. Chance is 0.250 throughout. The **distractor family is the independent variable**; the item, the instruction, k , and per-item option-order randomisation (seeded by item id) are held fixed across families, so a paired test across families is a test on byte-identical questions.

4.2 Distractor families

D-RAND draws other idioms’ figurative glosses at random. D-SURF draws glosses that share surface tokens with the gold. D-SEM draws glosses from idioms sharing semantic tags. D-LIT replaces one distractor with the idiom’s *own* expert literal gloss. D-HARD combines all three pressures.

4.3 Literal Attraction Rate

For the families containing a literal gloss, let ℓ_i be the literal option and g_i the gold:

$$\text{LAR} = \frac{1}{N} \sum_{i=1}^N \mathbf{1} \left[\arg \max_o p_i(o) = \ell_i \right].$$

LAR is not $1 - \text{accuracy}$: it isolates the share of errors that are the specific error of reading the idiom literally, and it has no floor, since a model that never reads literally scores 0 while still being wrong.

4.4 Seven interventions

E1 is the distractor ladder above. E2 swaps the idiom for a certified synonymous surface form. E3 perturbs the idiom as a unit (shuffle, reverse, delete one word, substitute one word, embed in its carrier sentence) with the options held fixed. E4 swaps the task language. E5 measures calibration and selective prediction. E6 ablates the option text itself (truncate, scramble, translate, strip to bare labels). E7 runs all $4! = 24$ orderings of a fixed option set.

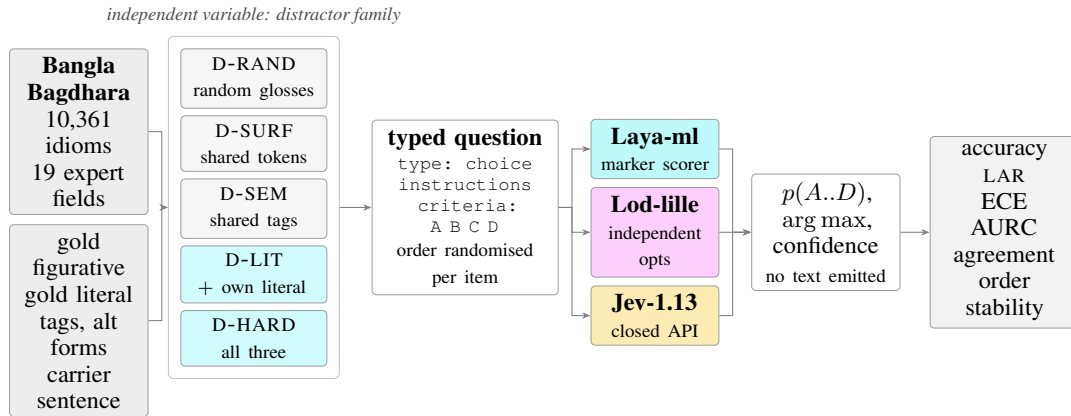


Figure 2: The instrument in one view. The corpus supplies both a gold figurative gloss and a gold *literal* gloss for each idiom, so the literal-bearing families (D-LIT and D-HARD) are assembled from expert annotation rather than written by us. Only the distractor family changes across conditions: the idiom, the instruction, $k = 4$ and the per-item order seed are held fixed, so any accuracy difference between two families on the same item is caused by the options alone. All three models receive the identical typed question (Laya-ml, Lod-lille, Jev-1.13) and return a probability over the four option keys and nothing else, which is the surface the audit works with.

4.5 Statistics and pairing

All intervals are 95% percentile bootstrap over items (2,000 resamples). Paired comparisons use McNemar’s exact test. ECE uses 15 equal-width bins; we also report AURC against a random-gate baseline.

Pairing verification. The two open arms were run in separate accounts from independent notebooks, each reconstructing the corpus and the task instrument from a hardcoded contract. On all 4,440 overlapping rows the gold key, gold position, idiom string and literal-distractor key agree at 1.000. Every paired test below is therefore on identical questions, not on re-derivations that happen to look alike.

4.6 Models

We audit three decision models spanning the design space (Table 1). **Laya-ml** is a bidirectional encoder with an option-marker scorer: all options are encoded together and scored from marker positions. **Lod-lille** is a decoder backbone with independent-option scoring, where each option attends only to the shared state, the question and itself from the same start position, order-independent by construction in fp32. **Jev-1.13** is a closed commercial model behind a decisions API; we know its interface and nothing else. It is not multilingual by documentation, which we exploit rather than regret: it provides the capability contrast that makes §5.1 a dissociation rather than an anecdote.

	Laya-ml	Lod-lille	Jev-1.13
backbone	mmBERT	Qwen3-0.6B	closed
scoring	marker	independent	unknown
params	322M	0.6B	n/a
items	8,876	888	8,761
latency	27.2 ms	86.1 ms	API
weights	open	open	closed

Table 1: The three arms. Cell tint identifies the model throughout the paper: cyan Laya-ml, magenta Lod-lille, yellow Jev-1.13. Lod-lille ran a fixed 10% random subset; Appendix I bounds the resulting drift at 0.016.

5 Analysis

5.1 Literal capture, and a crossover between the two open arms

Table 2 and Figure 3 give the ladder for the two open arms on the 888 shared items. Lod-lille is the stronger model on every non-literal condition, by up to +0.133. Insert the idiom’s own literal gloss and the ordering *reverses*: Lod-lille falls to 0.066, less than half of Laya-ml’s already-below-chance 0.151, and barely a quarter of chance. The same reversal holds for D-HARD.

Within this pair, then, LAR measures something accuracy does not: a model can be selected as “better” on a conventional benchmark while being worse at what that benchmark proxies for. Laya-ml and Lod-lille answer byte-identical questions in the same language, so the crossover is a paired result; whether it extends beyond this pair is not something two models can settle. The error is not diffuse: the decomposition under D-LIT is gold /

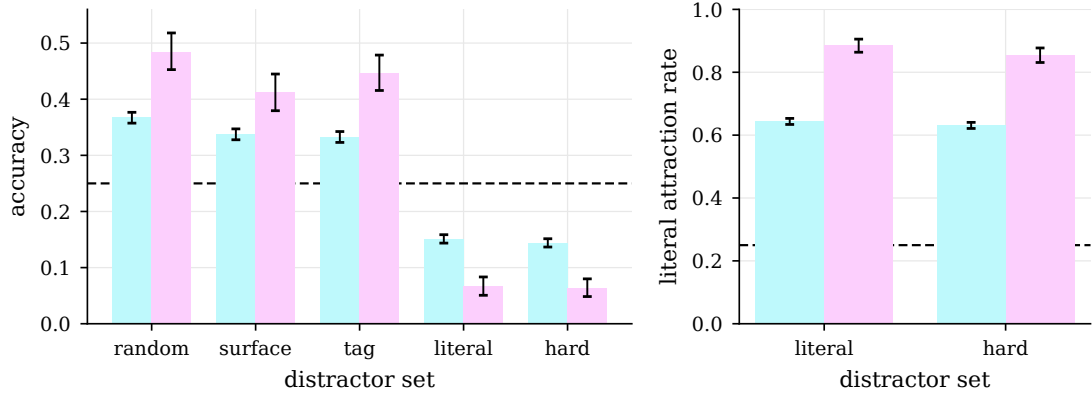


Figure 3: The distractor ladder (left) and literal attraction rate (right) for the two open arms. Laya-ml ($n = 8,876$), Lod-lille ($n = 888$); black dashed line is chance (0.250), error bars 95% bootstrap CIs. Surface- and tag-matched distractors cost a few points. Inserting the idiom’s own literal gloss (D-LIT) collapses both models far below chance, and collapses the *stronger* model harder: Lod-lille leads Laya-ml by +0.133 on random distractors and trails it by -0.070 once the literal gloss is present. On the right, both models prefer the literal reading to the figurative one on the majority of items.

Distractors	Laya	Lod	Δ
random	0.367	0.484	+0.133***
surface	0.338	0.412	+0.073***
tag	0.333	0.446	+0.120***
+ literal	0.151	0.066	-0.070 ***
hard	0.144	0.063	-0.086 ***
LAR (lit.)	0.644	0.885	
LAR (hard)	0.631	0.855	
mean p (lit.)	0.464	0.726	
mean p (gold)	0.195	0.128	

Table 2: The distractor ladder and literal attraction, McNemar on the 888 shared items (** $p < 0.001$). Bold marks the higher value per row, which is the better arm on the accuracy rows and the more captured arm on the LAR and p (lit.) rows. The sign of Δ flips at the horizontal rule: capability on the surrounding task does not protect against literal capture, and here is associated with more of it.

Model	n	Accuracy	95% CI
Laya-ml	8,876	0.367	[.357, .377]
Lod-lille	888	0.484	[.453, .518]
Jev-1.13	8,761	0.736	[.727, .745]

Table 3: The only condition identical across all three arms: Bangla idiom $\rightarrow$ Bangla gloss, random distractors. All pairwise McNemar tests on the overlapping items are significant: Jev–Laya +0.369, Jev–Lod +0.259, Lod–Laya +0.133 (all $p < 10^{-9}$).

literal / other = 0.151 / 0.644 / 0.207 for Laya-ml and 0.066 / 0.885 / 0.048 for Lod-lille, and on the clearest items Laya-ml places $p = 1.000$ on the literal gloss.

A confound we do not hide. Jev-1.13 reaches LAR = 0.047, two orders of magnitude below the open arms, but that number is *English*: the shared literal pool exists only for items with an English literal gloss, and the commercial arm ran it in English. Figure 6 draws the language $\times$ model matrix with the two unrun cells marked, so the missing experiment is visible rather than inferred. The Limitations give the full accounting.

The one matched three-way comparison. Exactly one condition is identical across all three arms: Bangla idiom, Bangla gloss, random distractors. Figure 8 and Table 3 report it. The commercial model is far ahead (0.736 against 0.367 and 0.484), which is the honest capability ordering, and the one against which the calibration results of §5.2 and §5.3 should be read.

5.2 Confidence is anti-diagnostic

Table 4 gives the mean probability assigned to the chosen option, split by outcome, on the literal-bearing families. Both open models are *more confident when captured than when correct*, Lod-lille extremely so (0.787 against 0.453). The reliability curves in Figure 4 slope *downward* under D-LIT in both.

Selective prediction therefore inverts. Restricting Laya-ml’s D-LIT to the most confident 5% of items gives 0.007 accuracy against 0.151 at full coverage, and the confidence-gated AURC is worse than a random gate for both open arms. The failure is not an artifact of one calibration philoso-

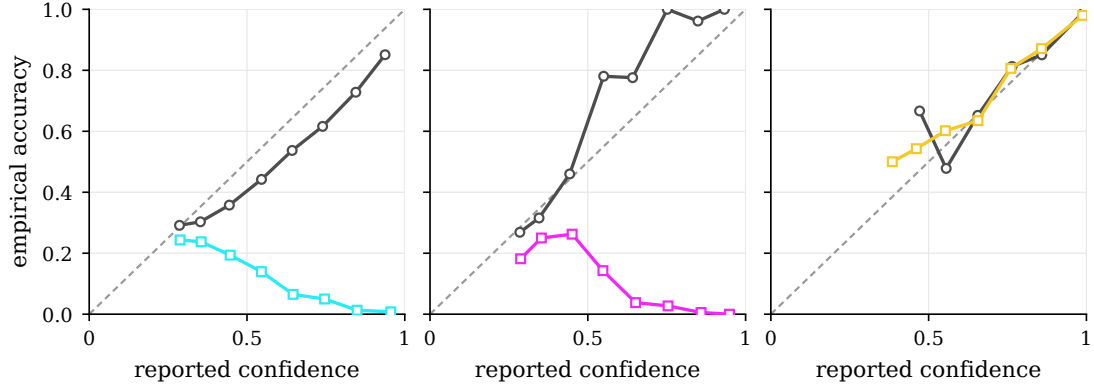


Figure 4: Reliability, 15 equal-width bins. Panels left to right: Laya-ml, Lod-lille, Jev-1.13. ■ circles are the random-distractor condition; squares in each panel’s own colour are the +literal condition. The grey dashed diagonal is perfect calibration. Under D-LIT the two open models’ curves bend *downward*, higher stated confidence buying lower accuracy, while Jev-1.13, running in English, stays on the diagonal. Reading the left two panels: the model is most wrong exactly where it reports being most sure.

Outcome	Laya	Lod	Jev
correct	0.432	0.453	0.935
captured	0.596	0.787	0.737
other error	0.421	0.399	0.746
ECE (D-RAND)	0.071	0.073	0.005
ECE (D-LIT)	0.385	0.679	0.014
AURC (conf.)	0.924	0.989	0.013
AURC (random)	0.849	0.933	0.069

Table 4: Confidence by outcome on the literal-bearing families, and calibration. Bold marks, per model, whether the model is more confident when correct or when captured. Both open arms are more confident when captured; their confidence-gated AURC is *worse* than the random-gate baseline in the grey row, which inverts selective prediction. Jev-1.13’s English condition does not show the inversion.

phy: Laya-ml ships fitted per-bucket temperatures, Lod-lille deliberately ships none, and Lod-lille further ships a *separate confidence head* whose documented purpose is estimating whether the answer is in the option list at all. The head does not rescue it: mean head confidence is 0.631 on captured items against 0.295 on correct ones, the same inversion.

Jev-1.13 is the interesting exception, and it is only an exception in one direction. Its reliability in English is close to ideal on the two families in Table 4 (ECE 0.005 and 0.014, rising to 0.047 on D-HARD), and its confidence-gated AURC beats random by an order of magnitude. That makes §5.3 harder to dismiss, not easier.

This does not contradict Ibrahim and Zaki (2026), who find decision-model calibration better than most LLM baselines; we reproduce that for the commercial arm on its own language. The narrower

claim is that in-distribution calibration need not survive a targeted distractor shift, and that it fails in the costly direction: a confidence-gated cascade (Li et al., 2026) accepts the high-confidence tail without review, and on the literal-bearing families that tail is where the open arms are *least* accurate.

5.3 A semantics-free control, on all three arms

The O4 ablation replaces every option’s text with a bare label. No semantic signal remains, so accuracy is pinned at chance by construction and whatever confidence survives is overconfidence by definition, with no modelling assumption needed to read it. This is the only condition in the paper that runs identically on all three arms: it requires no gloss, no literal annotation and no shared language, so the language confound that scopes §5.1–§5.2 to the two open arms does not arise here. Figure 5 reports it.

The ordering reverses against Table 4. The best-calibrated model under real options is the worst when the options carry nothing: Jev-1.13 reports 0.815 mean confidence at 0.254 accuracy, an ECE of 0.561. Lod-lille, the weakest of the three on accuracy, is the only one whose reported confidence collapses appropriately (0.282). We read this as the most portable finding in the paper, because it is the one that needs no annotation and no matched language to reproduce: a calibration number measured on in-distribution options says little about what a model will report when the option list is degenerate, and a caller thresholding on a returned probability is exposed to that case. Confidence-based acceptance is the mechanism of the cascade designs pro-

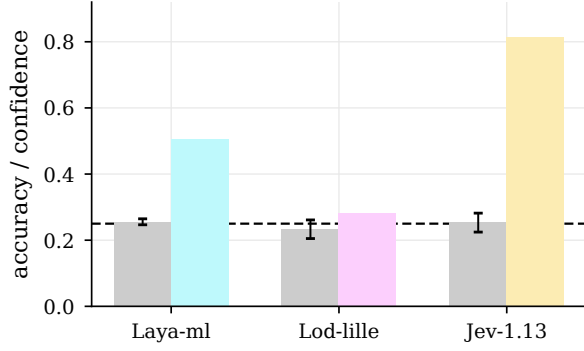


Figure 5: The semantics-free control. Per model, the grey bar is accuracy (with 95% CI) and the coloured bar is mean reported confidence (cyan Laya-ml, magenta Lod-lille, yellow Jev-1.13); black dashed line is chance. Because the options carry no information, accuracy is at chance by construction and the gap between the two bars is pure overconfidence: 0.250 for Laya-ml, 0.049 for Lod-lille, and 0.561 for Jev-1.13, which reports 0.815 confidence while answering at 0.254. Jev-1.13’s condition is English.

posed for this class (Li et al., 2026; Ibrahim and Zaki, 2026), and a degenerate option list is an ordinary consequence of an upstream bug or an empty retrieval rather than an adversarial construction.

A side-check comes with it: McNemar between the two open arms under O4 is not significant ($p = 0.76$), so the instrument’s null behaves identically in both.

5.4 Option order is architecture-specific

We run every item under all $4! = 24$ orderings of a fixed option set and record the identity of the chosen gloss, recovered as $\text{perm}[\text{slot}(\hat{y})]$.²

Table 5 and Figure 12 give a clean dissociation: a $7.8\times$ difference in answer stability and a $12\times$ difference in probability stability, tracking the scoring architecture rather than model size, backbone or capability. Lod-lille’s design claim, that each option attends only to the shared state, its question and itself from the same start position, is very nearly borne out. Laya-ml’s option-marker scorer is not, confirming an upstream issue report at scale.

The residual positional bias follows (Figure 11). With the gold placed uniformly, Laya-ml selects slot B 37.2% of the time against 14.6% for slot A, and its accuracy is 0.152 when the gold sits in

²An earlier version of this analysis counted distinct option keys (slot letters) and reported near-total instability for Laya-ml. That metric is an artifact: a key necessarily changes when the same gloss moves slots. All numbers here use the corrected gloss-identity metric; Laya-ml’s figure drops from 1.000 to 0.837 and the finding stands.

	Laya	Lod	Jev
answer changes	0.837	0.107	0.133
correctness varies	0.686	0.070	0.120
distinct glosses	2.83	1.14	1.15
perfectly stable	0.163	0.893	0.867
$p(\text{gold})$ range	0.346	0.029	0.128
selection spread	0.226	0.052	0.072

Table 5: Option-order sensitivity over all 24 orderings of the same four options ($n = 992 / 299 / 300$ items). Bold marks the least order-stable arm per row. The marker-based scorer changes its chosen gloss on 83.7% of items; the independent-option scorer, which is order-independent by construction in fp32, changes it on 10.7% and very nearly delivers what its design claims.

A against 0.403 in B, a $2.6\times$ swing from position alone.

Methodological consequence. Any evaluation of this class that fixes option order is partly measuring position. All results here randomise order per item, seeded by item id, and are averages over position rather than measurements at one of them; on the evidence here we would treat randomising order as a default for this class rather than as a robustness check.

5.5 Options are read as a bag of words

The O-series ablates the option text while holding the idiom fixed (Figure 10). Two readings matter. First, truncating each option to six words and *scrambling the words inside it* cost almost nothing: Laya-ml retains 0.967 and 0.925 of its signal, Lod-lille 0.909 and 0.948, Jev-1.13 is flat to within noise. Whatever all three models are doing on the option side is close to order-free lexical matching. Second, the drop to the grey semantics-free bar is the only large one in the figure, which is what certifies the instrument: the task really is measuring option semantics, it is just measuring them as a bag.

This converges with Sun et al. (2026), who permute which option name is bound to which definition and find the decision follows the name. We hold names fixed and destroy the text’s internal structure, and find the decision barely moves. Both point the same way: within an option these models read lexical identity rather than composed meaning.

5.6 Neither open model treats the idiom as a unit

E3 perturbs the idiom while holding the option list fixed (Figure 9). This is the experiment that replaces token attribution: since an idiom has no word-level meaning to attribute to, we instead ask whether the model behaves as though it stores the expression as a unit. A model that did would collapse when the unit is broken.

Neither does. Word-shuffling an idiom leaves Laya-ml’s accuracy statistically unchanged ($0.158 \rightarrow 0.168$) and reversal actually helps slightly (0.180); Lod-lille moves within noise on its 180 items. Answer agreement with the intact form under shuffling is 0.742 for Laya-ml and 0.917 for Lod-lille, so Lod-lille is more *stable*, consistent with §5.4, while being no more *compositional*.

The one manipulation that clearly helps both is embedding the idiom in its gold carrier sentence: $+0.080$ and $+0.117$. Context, not the idiom string, carries the usable signal, which is itself the diagnosis. These models are not retrieving a stored figurative entry; they are matching whatever lexical material is available, and a carrier sentence simply supplies more of it.

That cuts both ways: Xu (2026) flip 61.4% of initially correct decisions with short, fluent, task-preserving context. Our gold carrier sentences are the benign case of the same mechanism, and nothing in the interface distinguishes the two.

5.7 For the open arms, the failure localises to Bangla

Holding the model fixed and swapping only the task language (Figure 7) moves every arm by a large margin: Laya-ml $+0.246$, Lod-lille $+0.380$, Jev- $1.13 +0.233$ (all McNemar $p < 0.001$). Lod-lille reaches 0.864 in English against 0.484 in Bangla, so the figurative concepts are largely present; it simply cannot reach them through Bangla. Its documentation states that training evaluation was English-only, and this is what that looks like from outside the model.

Laya-ml’s control arm supports this reading: its English-only checkpoint run on Bengali script sits at 0.283 , barely above chance, which reproduces a documented failure mode and makes it unlikely that the Bangla result is an artifact of the scorer rather than of the language.

The qualification in the heading matters. Jev- 1.13 reaches 0.736 on the Bangla task (Table 3)

against 0.367 and 0.484 for the two models documented as multilingual, so a model with no multilingual claim outscores both. The deficit is a property of these two checkpoints rather than of the task or the script, which suggests the Bangla gap is reachable rather than intrinsic.

6 Discussion

Our results sit at three scopes and are worth keeping apart. The semantics-free control is the only one that runs identically on all three arms and needs no annotation or shared language, so it is the one we would expect to transfer. Literal capture and the confidence inversion are paired results on two architecturally unrelated open models answering byte-identical questions; replication across an encoder and a decoder scorer is real evidence, but two models are not a class. Option-order dependence tracks the scoring architecture rather than the model, and is the one place our audit confirms a design claim rather than denting one. The commercial arm escapes the first two in English and fails the third most severely, and we did not run the conditions that would say whether its resistance to capture belongs to the model or to the language it was tested in.

For practitioners we would draw three cautions rather than recommendations: a returned probability is worth little without a measurement of what the model returns on a degenerate option list; a benchmark score is not by itself evidence against literal capture, since on our two open arms the two move in opposite directions; and an evaluation that fixes option order is partly reporting position.

This is a boundary condition rather than a rebuttal: the cost and latency advantages are established (Tang and Zheng, 2026; Rao and Callison-Burch, 2026), and what we add is that the property the cascade designs depend on degrades first and silently. The typed interface guarantees a well-formed answer and nothing about whether it was arrived at for a reason, so each of these failures is invisible to a caller who only checks that the output validates.

7 Conclusion

We audited three decision models on Bangla idioms using only interventions on the option set, a method that needs no annotation and stays available when a model emits no text and its weights are closed. The result we would put most weight on is the cheapest to reproduce: with the options stripped of

semantics so accuracy is pinned at chance, the three arms report between 0.28 and 0.82 confidence, the arm best calibrated on real options returning 0.815 at 0.254. On the two open arms an idiom’s own literal gloss drives both well below chance, the two trade places, and confidence inverts. Option-order sensitivity separates the scoring architectures rather than the models. Whether capture behaves this way beyond this pair we cannot say, and §5.1 marks the conditions that would settle it.

Limitations

All three models are evaluated zero-shot, and this is the limitation we would weigh most heavily after the one below. Decision models are trained on a particular set of decision tasks, and the open arms’ own documentation reports that base checkpoints are weak on tasks outside that set. Some part of what we measure may therefore be out-of-distribution task transfer rather than literal capture specifically. The O4 control bounds this from one side, since it shows the instrument’s null behaving identically in both open arms, and the D-RAND to D-LIT contrast is within-model and within-item, which removes any fixed task-transfer offset. Neither argument rules the confound out. A single lightly fine-tuned arm would settle it, and we have not run one.

Laya-ml’s run had its per-bucket temperatures unset, so its ECE figures are raw. Lod-lille ships no temperature by design, so both open arms are effectively untempered and comparable on that axis; Jev-1.13’s internal calibration is unknown to us.

Lod-lille ran a fixed 10% subset (888 items), so its intervals are roughly ± 3 points and its stratified cells are thin. Appendix I bounds the subset bias at 0.016 and shows it changes no sign or ordering. E3 on the Lod-lille arm is 180 items and several of its perturbation contrasts are not individually significant. E7 covers 992 / 299 / 300 items rather than the full sets.

The cross-model LAR comparison in §5.1 is confounded with language. Jev-1.13 ran the literal condition in English while the open arms ran it in Bangla, and we did not run either of the two cells that would separate the model from the language: Jev-1.13 on the Bangla literal condition, and the open arms on the English one. Figure 6 draws both as unmeasured. This is why the capture and confidence claims are stated for the two open arms rather than for all three, and why §5.1 reports

a crossover between a specific pair rather than a trend in capability. The shared literal pool is also not a neutral sample of the corpus: in-pool LAR is higher than out-of-pool for Laya-ml while D-RAND accuracy is unchanged.

More broadly, this is one corpus, one language, two small open models and one commercial endpoint. The semantics-free control is the only result here that we would expect to transfer as stated; the rest should be read as findings about these systems on this material, and as an instrument other people can point at theirs.

Distractors are drawn from the same corpus, so difficulty is corpus-relative rather than absolute. Jev-1.13 is a commercial endpoint and may change without notice; our results are a snapshot, and we report the exact call count and cost in Appendix L so the snapshot is at least auditable.

Finally, no figure in this paper contains Bangla glyphs: the plotting toolchain available to us could not shape Bengali script. All categories are named in Latin script and qualitative Bangla examples are confined to the text.

Ethics Statement

The corpus is a published, expert-annotated resource of idioms (Sakhawat et al., 2026); it contains no personal data and no content about identifiable individuals. Our interventions rewrite option lists and perturb idiom strings, and produce no new annotation and no generated text.

The work audits two openly released models and one commercial API. We report failure modes in all three, including the vendor’s. We have taken care to separate what is measured from what is inferred. In particular, Figure 6 marks the cells we did not run rather than letting a favourable contrast stand unqualified, and §5.1 states the pool selection effect against our own cleanest number. Commercial endpoints are mutable; readers should treat the Jev-1.13 results as dated rather than as a standing property of the product.

Bangla is under-served relative to its speaker population, and we think an audit that locates a failure in the language rather than in the task is the more useful kind to publish. The instrument, the per-item outputs and the analysis code are released; Appendix L is the run record they correspond to.

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Request. For the commercial arm the body posted to `/api/alpha/decisions` is

```
{
  "model": "typesafe/jev-1.13",
  "state": <idiom, or carrier sentence>,
  "questions": {
    "q": {
      "type": "choice",
      "instructions": <one sentence, Table 6>,
      "criteria": {
        "A": <gloss>,
        "B": <gloss>,
        "C": <gloss>,
        "D": <gloss>
      }
    }
  }
}
```

The two open arms are called locally with the same four fields: the idiom as state, the instruction string, and the four option texts keyed A–D. The option order is shuffled per item by a generator seeded with the item id and the experiment seed, so the same item receives the same permutation in every arm and every family.

Response.

```
{
  "type": "choice",
  "choice": "B",
  "confidence": 0.75,
  "probabilities": {
    "A": 0.00,
    "B": 0.84,
    "C": 0.16,
    "D": 0.00
  }
}
```

We assert on every call that the returned keys are a subset of $\{A, B, C, D\}$ and that the probabilities sum to 1 within 10^{-2} ; no call in any run violated either.

Two confidence signals, and which one we use.

Note that confidence is not the maximum probability: in the example above it is 0.75 while $\max_o p(o) = 0.84$. It is a separate head whose documented job is to estimate whether the correct answer is present in the option list at all. Lod-lille ships the same arrangement. We log it as `conf_head` and report it where it matters (§5.2), but every cross-model number in this paper uses the maximum probability, because it is the only confidence signal all three arms define identically. Where the two disagree the head is not the more diagnostic of the pair: on Lod-lille’s captured items mean head confidence is 0.631 against 0.295 on correct ones, the same inversion the top probability shows.

Answer types we did not use. The interface also accepts `score` (a bounded numeric judgement) and `boolean` questions. We use `choice` exclusively and deliberately. A `score` question would ask the model to rate a gloss in isolation, which destroys the property the instrument depends on: that the gold and the literal reading compete inside

a single forward pass, exactly as the psycholinguistic account describes (Cacciari and Tabossi, 1988; Titone and Connine, 1999). A `boolean` question would do the same and additionally reintroduce a yes/no label whose name carries meaning of its own, which is the failure Sun et al. (2026) document. Keeping every experiment in one answer type also means the five distractor families differ in nothing but their option text.

Arm	Instruction string
Laya-ml	Bangla, see note
Lod-lille	Bangla, identical to above
Jev-1.13	“What is the actual figurative meaning of this idiom?”

Table 6: The instruction is one sentence and is byte-identical across families, experiments and the two Bangla arms. The Bangla string is the direct equivalent of the English one; it is not reproduced here because the typesetting toolchain used for this paper cannot shape Bengali script. It is released verbatim in `configs/task.json`.

B What each experiment changes

Table 7 states, for each experiment, the one thing that varies and the things held fixed. Reading down the “varies” column follows the argument: each row isolates one candidate explanation for the failure measured in E1.

	Varies	Fixed	Seed
E1	distractors	idiom, instr., k	1000
E2	idiom surface	meaning, options	2000
E3	idiom form	options, instr.	3000
E4	task language	item, family	4000
E5	nothing (derived)	n/a	n/a
E6	option text	idiom, gold	6000
E7	option order	option set	7000

Table 7: The seven interventions. E5 issues no calls of its own; it is computed from the E1 outputs. Seeds are the Bangla arms’; the commercial arm uses a disjoint set (Appendix L) because its blocks were run in a different order.

E1, the ladder. D-RAND samples three other idioms’ figurative glosses uniformly. D-SURF samples glosses sharing at least one surface token with the gold. D-SEM samples glosses from idioms sharing at least one semantic tag. D-LIT replaces one sampled distractor with the idiom’s own expert literal gloss. D-HARD applies all three pressures at once. Duplicates are removed and the slate is

topped up from the random pool if fewer than k distinct options survive.

E3, perturbing the unit. E3 runs on the literal-bearing option set, so its accuracies are comparable to D-LIT rather than to D-RAND: on the same 1,778 items Laya-ml scores 0.176 under D-LIT and 0.353 under D-RAND, against 0.158 for the intact idiom here. v0 is the intact idiom. v1 shuffles its words, v2 reverses them, v3 deletes one word at random, v4 substitutes one word with a word sampled from another idiom, and v5 replaces the idiom with its gold carrier sentence. The option list is held byte-identical across all six, so the only difference reaching the model is the state string.

E6, ablating the options. o0 is the full gloss. o1 truncates every option to its first six words. o2 shuffles the words within each option. o3 replaces each gloss with its counterpart in the other language. o4 replaces every option with a bare label carrying no semantics, which pins accuracy at chance by construction and is the control of §5.3.

C Output distributions

The three arms do not merely differ in accuracy; they place their probability mass very differently, and the shape of that mass is what makes the confidence results of §5.2 possible. Table 8 summarises the returned vectors.

		mean	med.	>.95	ent.
Laya	D-RAND	0.438	0.408	0.001	0.892
	D-LIT	0.536	0.479	0.039	0.778
Lod	D-RAND	0.447	0.391	0.002	0.881
	D-LIT	0.745	0.819	0.171	0.497
Jev	D-RAND	0.967	1.000	0.852	0.068
	D-LIT	0.931	0.990	0.708	0.131

Table 8: Distribution of the returned maximum probability, and normalised entropy of the full vector (1 = uniform). The literal-bearing family makes both open arms *sharper*, not flatter: Lod-lille’s mean peak rises from 0.447 to 0.745 and the share of near-deterministic answers from 0.002 to 0.171, while its accuracy falls from 0.484 to 0.066. The model is not confused by the literal gloss; it is convinced by it.

That last point is worth stating separately, because it is the mechanism behind the inverted reliability curves. A model that found the literal option merely distracting would spread mass and lose accuracy together. Both open arms do the opposite: adding the literal gloss *concentrates* the distribution

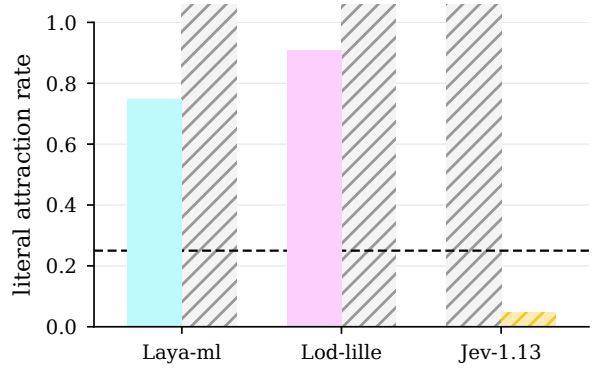


Figure 6: Literal attraction on the shared 3,265-item pool, by model and task language. Laya-ml, Lod-lille, Jev-1.13. Solid bars are Bangla; the hatched bar is Jev-1.13’s *English* condition, and the hatched blocks running past the top of the axis mark cells that were **never** run; they are not values of 1.0. Jev-1.13’s low attraction is therefore not comparable to the two Bangla values beside it, and the two unmeasured cells are the experiment that would make it so.

onto it. On 24 items Laya-ml returns $p = 1.000$ on the literal gloss and $p = 0.000$ on the gold.

Worked examples. Table 9 gives four of them. The idioms are transliterated because the toolchain cannot set Bengali script; the released data carries the original strings.

Idiom	Literal	Figurative (gold)	p_{lit}
tel	oil	arrogance	1.000
chamcha	spoon	sycophant	1.000
til	sesame seed	a tiny amount	1.000
charpaya	four-legged	rope cot	1.000

Table 9: Laya-ml under D-LIT. In each case the expert literal gloss and the expert figurative gloss are both in the option list, and the model places all of its mass on the literal reading. These are not hard or obscure idioms; three of the four are single words in everyday use.

D The language confound

Figure 6 draws literal attraction as a model $\times$ task language matrix on the shared pool, with the two cells we did not run marked as unmeasured. It is the basis for scoping the capture and confidence claims of §5.1 and §5.2 to the two open arms.

E Cross-lingual arms

Figure 7 plots the E4 conditions discussed in §5.7.

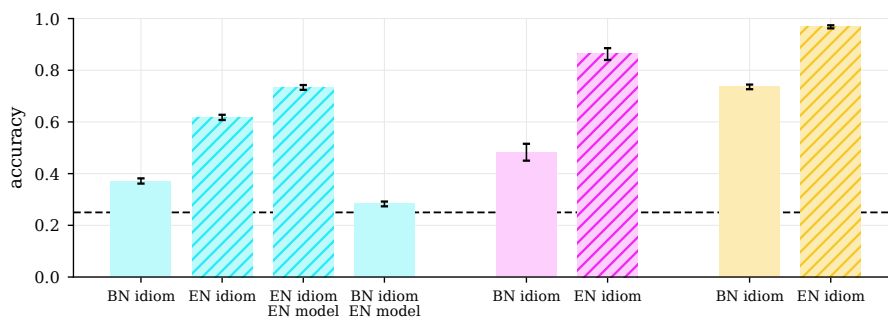


Figure 7: Accuracy by task language, grouped by model: ■ Laya-ml (four bars), ■ Lod-lille, ■ Jev-1.13. **Solid** bars are Bangla input, **hatched** bars English; Laya-ml’s third and fourth bars use the English-only checkpoint, the fourth being the control. Error bars 95% bootstrap CIs, dashed line chance. Every model gains 20–45 points from the same idiom presented in English.

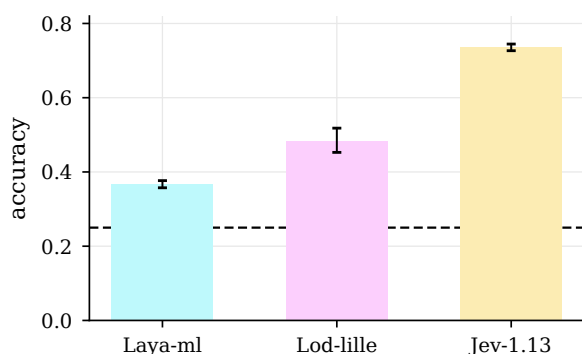


Figure 8: The matched three-way comparison. Bars left to right: ■ Laya-ml, ■ Lod-lille, ■ Jev-1.13. Black dashed line is chance, error bars 95% bootstrap CIs. This capability ordering is the baseline against which the calibration failures in §5.2–§5.3 should be read.

F The matched three-way condition

Figure 8 plots Table 3, the one condition that is identical across all three arms.

G Unit integrity

Figure 9 plots the E3 perturbations described in §5.6. The options are byte-identical across all six variants, so the only difference reaching the model is the state string.

H Surface invariance

The corpus ships alternative surface forms certified by annotators as meaning-preserving. A model reading meaning should be invariant across such a pair. Laya-ml agrees with itself on 0.657 of 3,143 pairs (2,423 items) and Lod-lille on 0.677 of 310 pairs (234 items): about a third of the time, a meaning-preserving rewrite changes the answer. Mean Jensen–Shannon divergence between the two probability vectors is 0.039 and 0.085.

I Subset representativeness

Lod-lille ran a fixed 10% random subset. Table 10 compares Laya-ml’s accuracy on those same 888 items against its own full 8,876.

Family	full	subset	drift
D-RAND	0.367	0.351	−0.016
D-SURF	0.338	0.339	+0.001
D-SEM	0.333	0.325	−0.007
D-LIT	0.151	0.136	−0.015
D-HARD	0.144	0.149	+0.004

Table 10: Laya-ml on the 10% subset against its own full set. Maximum drift 0.016. Two of five subset estimates fall marginally outside the full-set bootstrap CI, which is expected at $n = 888$ and changes no sign or ordering. Cite the full-set figures as the precise ones.

J Full option ablation

Table 11 gives the numbers behind Figure 10.

Option text	Laya	Lod	Jev
full gloss	0.365	0.492	0.970
truncated	0.361	0.468	0.958
scrambled	0.357	0.479	0.974
other lang.	0.347	0.398	0.963
no semantics	0.256	0.233	0.254

Table 11: Option-side ablation. The grey row is the semantics-free control, at chance by construction in all three arms. Jev-1.13’s column is English.

K Order sensitivity and position bias

Figure 11 gives the per-slot selection shares behind the summary in §5.4.

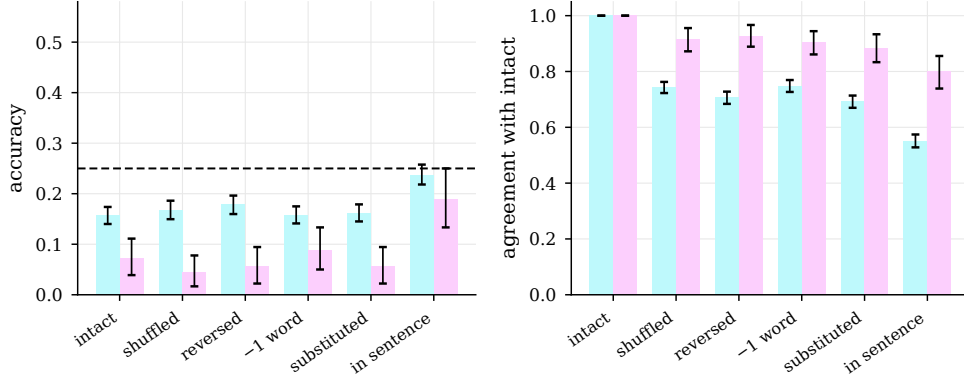


Figure 9: Unit integrity: the idiom is perturbed, the options held fixed. ■ Laya-ml, ■ Lod-lille; error bars 95% bootstrap CIs, dashed line chance (left panel). *Left*: accuracy. *Right*: agreement with the answer on the intact idiom. Scrambling and reversing a non-compositional expression does not meaningfully hurt either model, which is the bag-of-words signature.

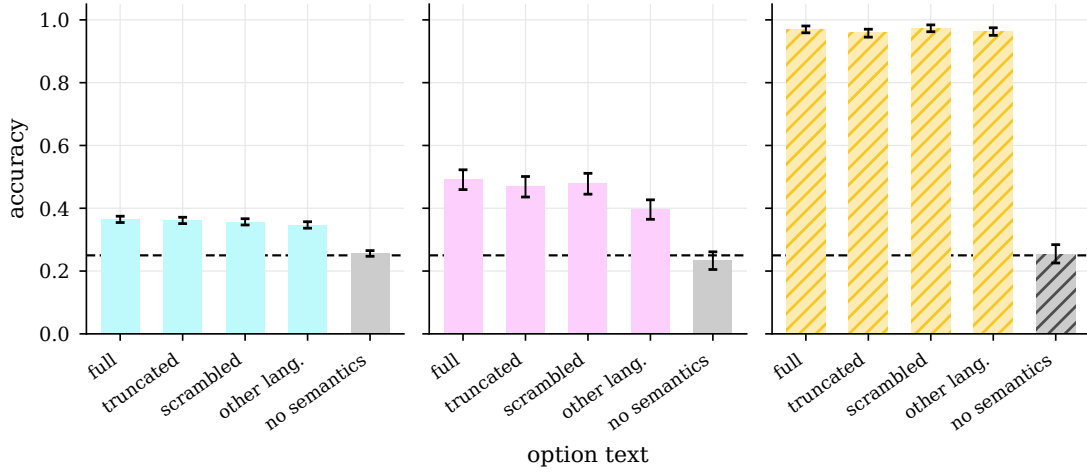


Figure 10: Option-side ablation. Panels left to right: ■ Laya-ml, ■ Lod-lille, ■ Jev-1.13. In each panel the fifth, ■ bar is the semantics-free control of §5.3; the whole Jev-1.13 panel is hatched because its task is in **English**. Error bars are 95% bootstrap CIs and the black dashed line is chance. Truncating and word-scrambling the option text cost almost nothing in any of the three models.

L Run record

Environment. Python 3.13.15 throughout. The two open arms ran on one Tesla T4 (15,360 MiB): Laya-ml in `bf16` with `max_len` 1024 at 27.2 ms per call, Lod-lille in `fp32` at 86.1 ms per call. The commercial arm needs no accelerator and ran on CPU against `typesafe/jev-1.13`.

Temperatures. Laya-ml’s run recorded `temperature = [1.0, 1.0, 1.0]` with `temperature_by_options` empty, so the per-bucket temperatures it ships were not applied and its ECE figures are raw. Lod-lille ships `temperature: 1.0` by design, its authors having reported that fitting hurt held-out calibration. Both open arms are therefore untempered and comparable on that axis. The commercial arm’s

internal calibration is not observable.

Seeds. The Bangla arms use a per-experiment seed (E1 1000, E2 2000, E3 3000, E4 4000, E6 6000, E7 7000) added to the item id, so each item’s option permutation is reproducible and identical across arms. The split seed is 20261001. The commercial arm uses a disjoint block seed set (order audit 17000, literal 12000, ceiling 11000, Bengali control 14000, options 16000) because its blocks ran in a different order; the permutation for a given item is still a pure function of seed and id.

Call volume and cost. The commercial arm issued 56,533 calls for \$0.9348 at \$0.042 per million input tokens with output free. The blocks are the order audit ($300 \times 24 = 7,200$), literal capture on the shared pool ($3,265 \times 3 = 9,795$), the English

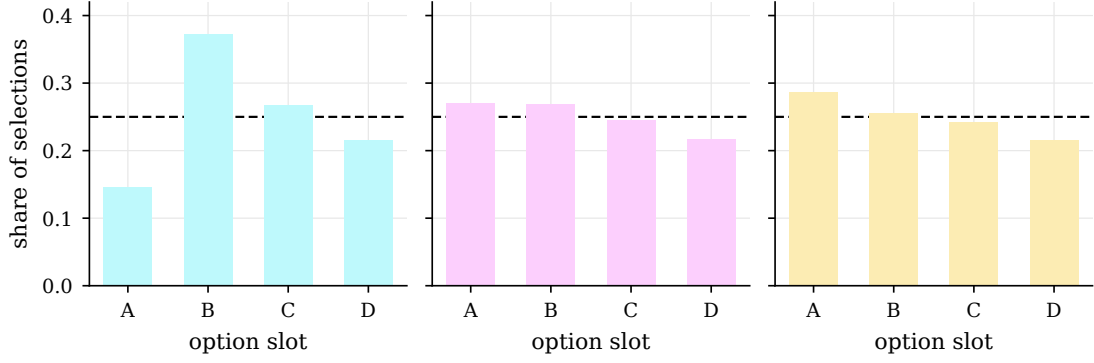


Figure 11: Share of selections falling on each option slot. Panels left to right: Laya-ml, Lod-lille, Jev-1.13. The gold option is placed uniformly at random, so the black dashed line at 0.25 is the expectation under no positional bias and any departure from it is bias. Laya-ml takes slot B 37.2% of the time and slot A 14.6%; the other two show only a mild primacy gradient.

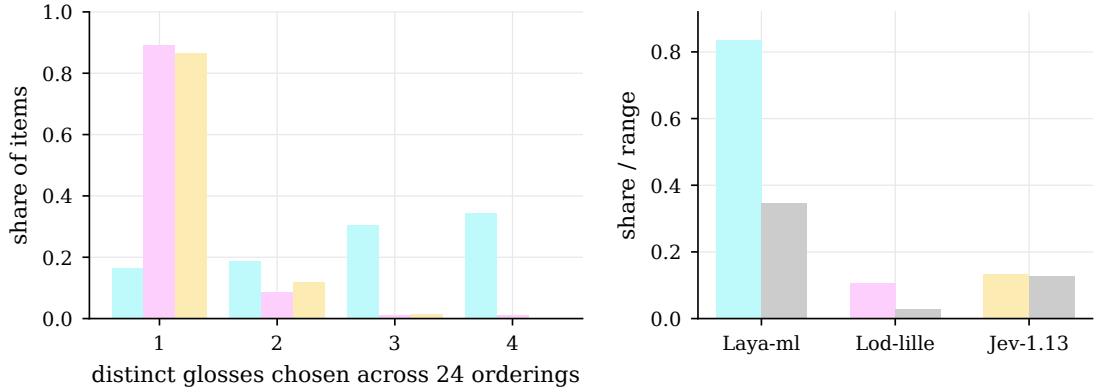


Figure 12: Order sensitivity. *Left*: how many distinct glosses each model selects across the 24 orderings of one option set; within each group bars run Laya-ml, Lod-lille, Jev-1.13, and a value of 1 means perfect stability. *Right*: per model, the coloured bar is the share of items whose chosen gloss is unstable and the grey bar is the mean within-item range of $p(\text{gold})$. Laya-ml is unstable on 83.7% of items with a mean $p(\text{gold})$ swing of 0.346; the other two are near-stable.

ceiling ($8,761 \times 3 = 26,283$), the Bengali control (8,761) and the option ablation ($873 \times 5 = 4,365$), which is 56,404 retained rows. The remaining 129 calls are the endpoint probe and the response-contract checks made before each block. Transient 429 and 5xx responses were retried with exponential backoff to a 30 s ceiling; no call was dropped.

Statistics. All intervals are 95% percentile bootstrap over items, 2,000 resamples, fixed seed 0. ECE uses 15 equal-width bins. Paired comparisons use McNemar’s exact test via the binomial distribution. AURC is reported against a random-gate baseline computed on the same items.

What is released. The per-item outputs for every cell of every experiment (one row per item per condition, carrying the full probability vector, the predicted key, the gold key, the literal key and the gold position), the five split manifests, the three

run environment records quoted above, the task configurations containing the verbatim instruction strings, the analysis scripts that turn those outputs into every table in this paper, the figure sources, and a CSV of the exact plotted values beside each figure.

Pairing. Because the two open arms were reconstructed independently from a written contract rather than from shared code, the released outputs admit a direct check: on the 4,440 overlapping rows the gold key, gold position, idiom string and literal-distractor key agree at 1.000. That check is itself a released artifact, and it is what licenses every paired test in the paper.

Known gaps. E2 was not run on the commercial arm, because the alternative surface forms exist only in Bangla and the arm’s task is English. The two unrun cells of Figure 6 are the Bangla literal

1072 condition for the commercial arm and the English
1073 literal condition for the open arms. Neither was
1074 attempted, and neither is estimated anywhere in
1075 this paper.